



NUST

NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

P R O J E C T R E P O R T

Guardian Knot: A C++ Based Automated Persistence Safety System for Emergency Monitoring



Subject : Computer Programming

Submitted To : Sir Shah Muhammad

Submission Date : April 28th 2026

G R O U P M E M B E R

Student Name	CMS ID
Nimra Mumtaz	553839

1)Executive Summary:

The "Guardian Knot" is a human-centric security system designed using C++. Unlike traditional systems that only focus on unauthorized entry, this project addresses the "Safety Gap" by monitoring whether the user is conscious, safe, or being forced to act under duress. It uses a "Dead Man's Switch" logic to ensure continuous protection.

2)Problem Statement:

Current security systems have three major flaws that this project solves:

- **Safety Gap:** They don't know if the user is physically able to call for help.
- **Coercion Risk:** Intruders can force users to unlock systems.
- **Technical Noise:** Traditional logs are too technical for quick emergency response.

3) Key Features & Logic:

The system is built on four core technical pillars:

- **Heartbeat Timer:** Using the `time(0)` function, the system tracks the gap between user check-ins. If the user does not respond within 60 seconds, an automatic emergency is triggered.
- **Dual-PIN Security:** The system has two separate codes:
 1. **Normal PIN:** For safe access.
 2. **Silent Duress PIN:** If used, the system pretends to work but secretly logs a "Silent Cry for Help" and triggers a discreet alarm.
- **Interactive Feedback:** Using `windows.h`, the system provides visual (Color Coding: Red for Danger, Green for Safety) and auditory (2500Hz Beeps) feedback.
- **Emergency Logging:** Every event is recorded in `EMERGENCY_LOG.txt` with exact timestamps for medical or legal review.

4. Implementation Results:

- **Registration:** Setting up the user profile and PINs.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
GUARDIAN KNOT - SECURITY SYSTEM
=====
[NEW SYSTEM] Let's secure your life...
Enter Your Name: Nimra
Set Main Password: 1234
Set Emergency Duress PIN: 5678

[SUCCESS] You are now protected. Monitoring Active.
```

- **Safe Login:** Green screen confirmation showing "Safety Confirmed."

```
Set Emergency Duress PIN: 5678

[SUCCESS] You are now protected. Monitoring Active.
```

- **Alarm Trigger:** Red screen and Beep sound when the 60s timer expires.

```
=====
Welcome back, Nimra.

Enter Password/PIN to confirm you are safe: 1234

[!!!] DANGER: EMERGENCY ALERT ACTIVATED! [!!!]
```

- **Log File:** Showing the text inside EMERGENCY_LOG.txt.

```
|
--- CRITICAL ALERT: NO RESPONSE ---
Time of Event: Thu Apr 23 13:49:18 2026
Alert: Nimra has not checked in for 339 seconds.
Fear: We are deeply worried for their safety. Please check on them!
-----

--- SILENT CRY FOR HELP ---
Time of Event: Thu Apr 23 13:52:20 2026
Condition: Nimra is being forced to unlock the system.
Note: They cannot call for help. Please reach out to them now!
-----
```

5. Future Enhancements:

To make the system even more robust, we plan to:

1. Integrate **GSM Modules** to send automated SMS alerts to the police.
2. Use **Pulse Oximeter sensors** to trigger alerts if the user's heart rate becomes abnormal.
3. Develop a **Mobile App interface** for remote monitoring by family members.

6. Conclusion:

In the silence of an emergency, every second counts. This system is the voice for the unresponsive." This project successfully demonstrates how simple C++ logic can be used to save lives in real-world scenarios.
